ACS Electrochemistry

Open Access

This article is licensed under CC-BY 4.0

pubs.acs.org/electrochem

Tutorial

DRTtools: Freely Accessible Distribution of Relaxation Times Analysis for Electrochemical Impedance Spectroscopy

Zilong Wang,[#] Yuhao Wang,[*,#] Baptiste Py, Adeleke Maradesa, Jiapeng Liu, Ting Hei Wan, Mattia Saccoccio, and Francesco Ciucci[*]

Cite This: *ACS Electrochem.* 2025, 1, 2680−2689

Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

KEYWORDS: *Electrochemical impedance spectroscopy, distribution of relaxation times, charge-transfer kinetics, interfacial effects, fuel cells, batteries*

Downloaded via 103.12.174.89 on April 3, 2026 at 14:38:43 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

■ INTRODUCTION

Electrochemical impedance spectroscopy (EIS) has become an indispensable characterization technique across numerous scientific disciplines, from fundamental electrochemistry and battery research to corrosion science, electrocatalysis, and even biological sensing.[1−3] By applying a small alternating current (AC) perturbation and measuring the system's current response at various frequencies, EIS generates an impedance spectrum, providing a detailed fingerprint of the underlying processes. However, extracting meaningful physical parameters from this spectrum often presents a significant challenge. The conventional approach involves fitting experimental data to equivalent circuit models (ECMs), which are assemblies of resistors, capacitors, inductors, and other elements designed to represent specific electrochemical phenomena. While ECM regression is a ubiquitous technique, the development of an appropriate ECM for a given electrochemical system is often challenging and heavily relies on expert knowledge. Furthermore, the interpretation of the regression results can be ambiguous, as multiple distinct circuit configurations may provide equally good fits to the same impedance spectrum.[4,5] This ambiguity hinders rigorous analysis and can obscure understanding, often forcing researchers into subjective trial-and-error processes for the identification of suitable circuit models, which heavily rely on prior assumptions. Consequently, there is a pressing need for a simple, generalized, and less model-dependent approach to interpret EIS data effectively.

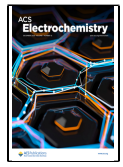
To address the limitations of equivalent circuit models, the distribution of relaxation times (DRT) method has emerged as a powerful and complementary tool for EIS analysis.[6−16] Unlike model-dependent fitting, DRT is a technique that transforms frequency-domain impedance data into a time-domain distribution of characteristic timescales. This approach conceptually represents the total system impedance as an infinite series of parallel resistor−capacitor (RC) elements, each defined by a unique time constant.[17−19] The deconvolution of the impedance spectrum yields a distribution function, often denoted as $\gamma(\log \tau)$. Under the separated, series configuration assumption, an individual peak corresponds to a distinct relaxation process. The peak's position on the logarithmic timescale axis indicates the characteristic time constant of the process, while its area is proportional to the associated resistance contribution. This time-domain representation offers the significant advantage of visualizing coupled or overlapping processes that might otherwise be convoluted semicircles in a conventional Nyquist plot, thereby providing direct visualization of specific phenomena such as charge-transfer kinetics and interfacial effects that are otherwise complex to analyze.

DRT has become a valuable tool for system diagnostics, performance evaluation, and fundamental scientific investigations. In the battery field, the DRT has been used to separate contributions from SEI/CEI formation, charge-transfer kinetics, and solid-state diffusion in solid-state electrolytes.[10,11,20−24] In reference to fuel cells, the DRT model has been used isolate membrane dehydration, anode poisoning, gas-transport limitations, and proton-transfer kinetics, clarifying degradation pathways.[12,25−30] Broader applications have also emerged in corrosion science, electrolyzer development and analysis, gas sensing, supercapacitors, and, even, agriculture, biology, and medicine.[31−41]

■ BASIC THEORY OF THE DISTRIBUTION OF RELAXATION TIMES METHOD

The foundation of EIS lies in treating the electrochemical system as a linear, time-invariant, and stable system when subjected to a small perturbation. The system's response at the frequency f is then characterized by its impedance, $Z(f) = V(f)/I(f)$, defined as the ratio of the Fourier transform of voltage response, $V(f)$, to the Fourier transform of the current

Received: August 4, 2025
Revised: October 10, 2025
Accepted: October 17, 2025
Published: November 5, 2025



© 2025 The Authors. Published by American Chemical Society

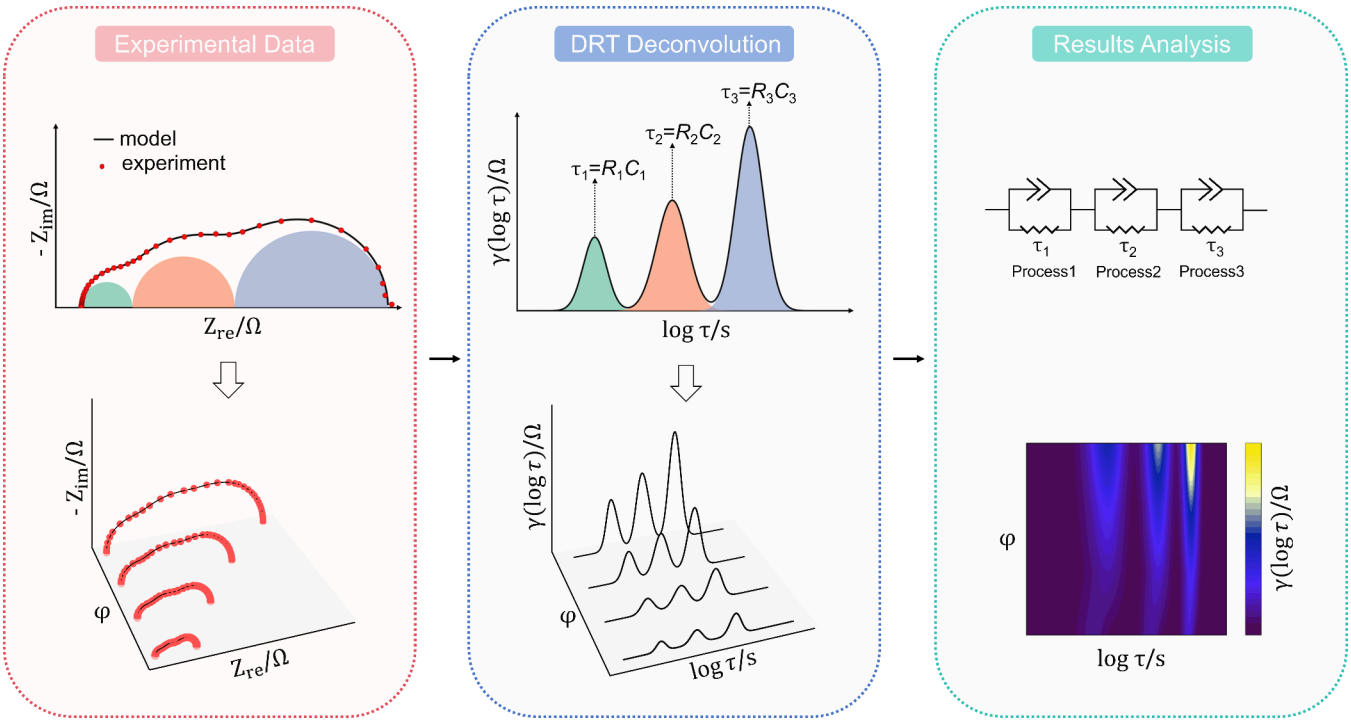


Figure 1. Schematic diagram of the conversion of one impedance spectrum (upper panel) and several spectra (lower panel) into different DRTs, followed by subsequent analysis to identify the associated electrochemical processes.

stimulus, $I(f)$.[42] This impedance function can be conceptualized as originating from a collection of physical processes, each characterized by a specific time constant τ. In the context of DRT, the total impedance of an electrochemical system is modeled as the sum of an Ohmic resistance (R_∞), an inductance ($i\omega L_0$) that accounts for high-frequency contributions of wiring etc., and a distribution of parallel RC elements, i.e.,

$$Z(f) = R_\infty + i\omega L_0 + \int_{-\infty}^{+\infty} \frac{\gamma(\log \tau)}{1 + i2\pi f\tau}\, d \log \tau \tag{1}$$

where $\gamma(\log \tau)$ is the DRT such that $\gamma(\log \tau)\, d \log \tau$ represents the contribution to the total resistance from processes with log-time constants between $\log \tau$ and $\log \tau + d \log \tau$. In most cases, the DRT is assumed to be non-negative. Furthermore, its integral over all timescales should ideally sum up to the total polarization resistance R_p, i.e., $\int_{-\infty}^{+\infty}\gamma(\log \tau)\, d \log \tau = R_p$. Equation 1 is commonly used in DRT analysis, where the peaks in the $\gamma(\log\tau)$ vs $\log\tau$ plot provide direct insights into the electrochemical processes: (1) peak position—the position of a peak on the $\log \tau$ axis indicates the characteristic time constant of a relaxation process, knowing that different processes (e.g., charge transfer) occur at distinct timescales, which leads to peaks at different positions; (2) peak area—the area under a peak ($\int_{\text{peak}}\gamma(\log \tau)\, d \log \tau$) corresponds to the resistance associated with that specific process, as illustrated in Figure 1.

Obtaining $\gamma(\log \tau)$ from the experimental impedance data $Z(f)$ involves solving the DRT integral equation shown in eq 1. However, since $Z(f)$ is only measured at a finite number of frequencies and is laden with experimental noise, this constitutes an ill-posed inverse problem.[43,44] Small errors or noise in the impedance data can lead to large, non-physical oscillations or instabilities in the resulting $\gamma(\log \tau)$ distribution. Consequently, directly minimizing the difference between the experimental data and the model prediction based on the integral equation (e.g., by simple least squares) does not lead to meaningful DRT recoveries.

To overcome the ill-posed nature of DRT deconvolution, and obtain a stable, smooth, and physically meaningful distribution of timescales, regularization techniques are employed.[45,46] Regularization introduces constraints or penalties on the solution space, biasing the result towards desirable properties such as smoothness or sparsity. The most common regularization method is Tikhonov regularization,[47] which modifies the standard least squares objective function by adding a penalty term, $\lambda\|\boldsymbol{Lx}\|^2$, where λ is the regularization parameter, $\boldsymbol{L}$ is the derivative/differentiation matrix and $\boldsymbol{x}$ is the vector of discretized DRT at $\log \tau$. The goal then becomes to minimize the sum of squared residuals (the difference between experimental and model impedances) completed by a term proportional to the "roughness" or "complexity" of the obtained $\gamma(\log \tau)$ so that the recovered DRT, $\gamma_{\text{opt}}(\log \tau)$, is obtained as

$$\gamma_{\text{opt}}(\log \tau) = \underset{x \geq 0}{\operatorname{argmin}}(\|\mathbf{A}\boldsymbol{x} - \mathbf{Z}_{\text{exp}}\|^2 + \lambda\|\boldsymbol{Lx}\|^2) \tag{2}$$

where $\mathbf{A}$ is the discretization matrix, $\mathbf{Z}_{\text{exp}}$ is the experimental impedance vector, and $\|\cdot\|$ is the Euclidean norm. The matrix $\boldsymbol{L}$ is typically chosen to penalize the magnitude of the first or second derivative of $\gamma(\log \tau)$ with respect to $\log \tau$, hence promoting smoothness in the deconvolved distribution, with higher-order derivative penalties resulting in stronger smoothness.[48]

■ DRTtools.com TUTORIAL

Our prior developments, including the MATLAB-based DRTtools and Python-based pyDRTtools,[6] have been instrumental in advancing DRT analysis. However, these

https://doi.org/10.1021/acselectrochem.5c00334
ACS Electrochem. 2025, 1, 2680−2689

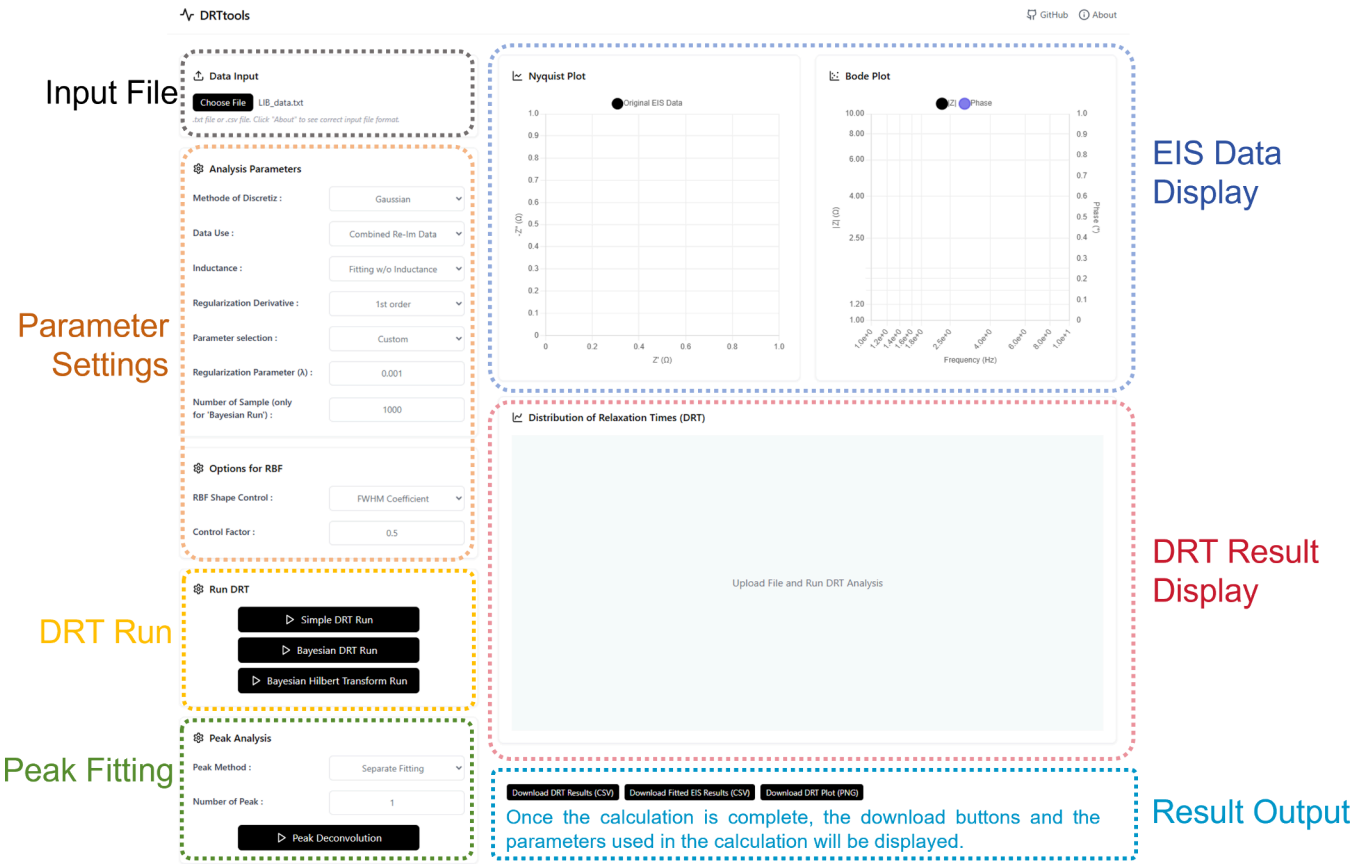


Figure 2. Screenshot of the DRTtools.com interface webpage separated according to the various functionalities offered.

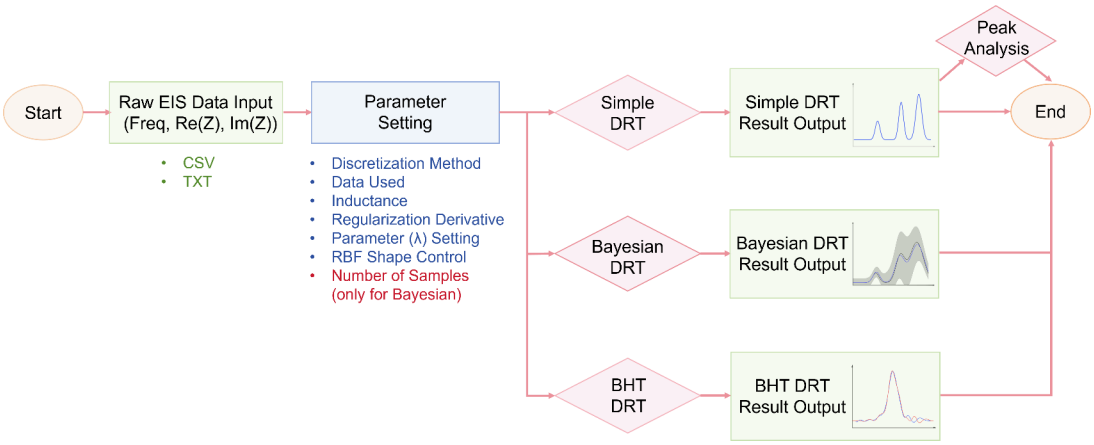


Figure 3. Flowchart of DRT analysis through DRTtools.com.

platforms imposed a usage threshold like software license requirements or the necessity for a certain level of programming proficiency, thereby limiting their widespread adoption. Recognizing this barrier, we developed DRTtools.com, a freely accessible web platform designed for instant use without any installation, offering core functionalities identical to MATLAB-based DRTtools and Python-based pyDRTtools versions, making it easily accessible and user-friendly even for those without programming experience, and globally accessible online (Figure 2).

DRTtools.com can perform comprehensive DRT analysis, including flexible regularization parameter selection, DRT deconvolution using different methods, probabilistic DRT analysis through Bayesian DRT analysis, data quality assessment through the Bayesian Hilbert Transform (BHT) DRT analysis, and peak analysis. This section provides a step-by-step tutorial to facilitate the use of this platform (Figure 3).

Data Input. The DRTtools.com interface is organized with input parameters on the left panel and interactive plots on the right. To begin an analysis, the user must first import experimental EIS data by clicking the "*Choose File*" button located in the "*Data Input*" panel. The platform supports both .csv and .txt file formats. The data within these files must be organized into three headerless columns corresponding to frequency, the real part of impedance (Z'), and the negative imaginary part of impedance ($-Z''$). It is strongly recom-

https://doi.org/10.1021/acselectrochem.5c00334
ACS Electrochem. 2025, 1, 2680−2689

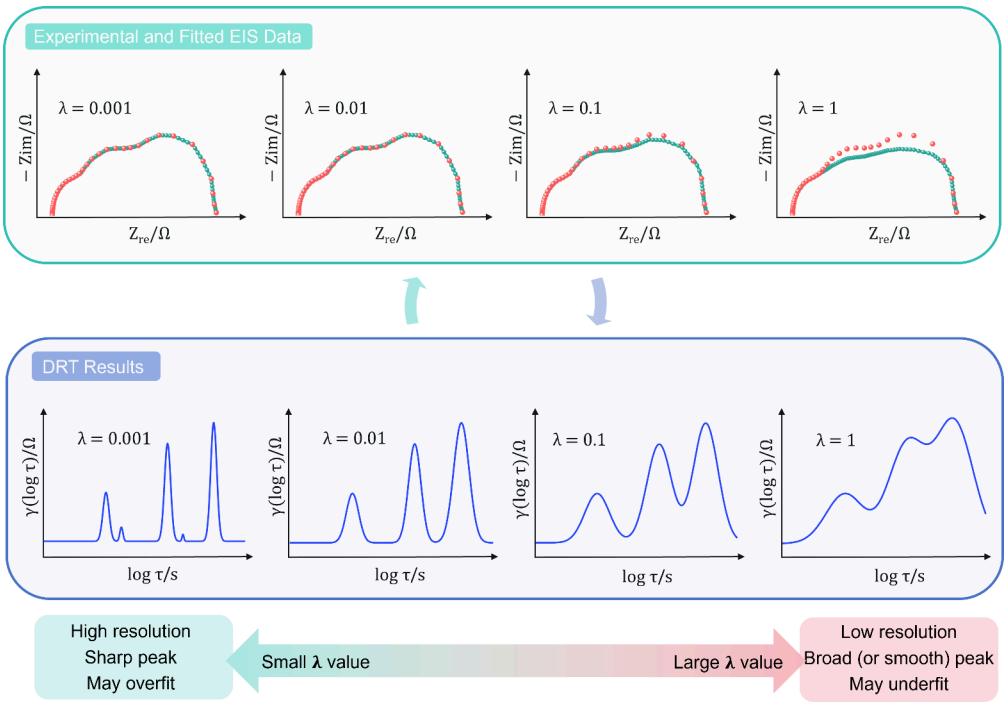


Figure 4. Recovered DRTs for different values of the regularization parameter λ.

mended to confirm the file is correctly formatted before uploading. Upon successful import, the main plot area will display both Nyquist and Bode plots of the experimental data.

Parameter Settings. After importing the data, the user proceeds to the "*Analysis Parameters*" and "*Options for RBF*" panels to configure the settings for the DRT deconvolution. This step is necessary to define the properties of the analysis and the radial basis functions before running the calculation.

Discretization Functions. Users must first select the basis function to approximate the DRT function. The default option is the "*Gaussian*" radial basis function (RBF),[49,50] but other choices are available, including C2, C4, and C6 Matérn, Inverse Quadratic, Inverse Multiquadric, and Cauchy functions. The selection of the RBF type directly influences the shape and smoothness of the final DRT result. Compared to piecewise linear (PWL) discretization, RBFs can provide faster numerical convergence and can be advantageous when the distribution decays smoothly near the edges of the frequency range (more details are provided in Figure S1 and Table S1 in the Supporting Information).

Data Use. Next, the user selects which part of the impedance data will be used for the analysis. The options are the real part ("*Re Data*"), the imaginary part ("*Im Data*"), or both ("*Combined Re-Im Data*"). The combined data setting is standard and recommended because it utilizes the full information content of the complex impedance.

Inductance. To address potential inductive features in the high-frequency regime, the user can choose one of three options. The analysis can proceed by fitting the data without an inductor ("*Fitting w/o Inductance*"), by including an inductor in the model ("*Fitting with Inductance*"), or by removing the high-frequency data containing inductive features altogether ("*Discard Inductive Data*"). Fitting with an inductor is suitable when these features are clearly visible in the Nyquist plot, whereas discarding the inductive data is the preferred method to completely remove their influence from the analysis (more details are provided in Figure S2 in the Supporting Information).

Regularization Derivative. The user must also select the differentiation order of the $\boldsymbol{L}$ matrix, with "*1st order*" and "*2nd order*" being the typical options. These choices penalize the magnitude of the first and second derivatives of the DRT, respectively. Using the second-order derivative is a common method to obtain smoother DRT peaks. The chosen derivative order directly influences how the regularization parameter (λ) regulates smoothness in the final result (more details are provided in Figure S3 in the Supporting Information).

Regularization Parameter Selection. The selection of the regularization parameter (λ) is a critical step that governs the quality of the deconvolved DRT. This parameter controls the fundamental trade-off between fitting the experimental data closely and ensuring the smoothness of the resulting distribution. As shown in Figure 4, a large λ value will produce a very smooth DRT but may obscure or distort true features (underfitting). In contrast, a small λ value can capture fine details but may also amplify noise and create non-physical peaks (overfitting).

- Large λ values prioritize smoothness by suppressing oscillations and noise but potentially oversmoothing the true features, merging peaks, or distorting their shapes (underfitting, Figure 4).
- Small λ values prioritize fitting the impedance data, thereby potentially capturing fine features but also amplifying noise, which often leads to non-physical oscillations and spurious peaks (overfitting, Figure 4).

By default, the λ value is set to 0.001. However, for more accurate deconvolution and to achieve optimal results tailored to specific datasets, this parameter should ideally be fine-tuned. Finding an optimal λ that balances this bias−variance trade-off is a central challenge. The platform provides two approaches for this:

https://doi.org/10.1021/acselectrochem.5c00334
ACS Electrochem. 2025, 1, 2680−2689

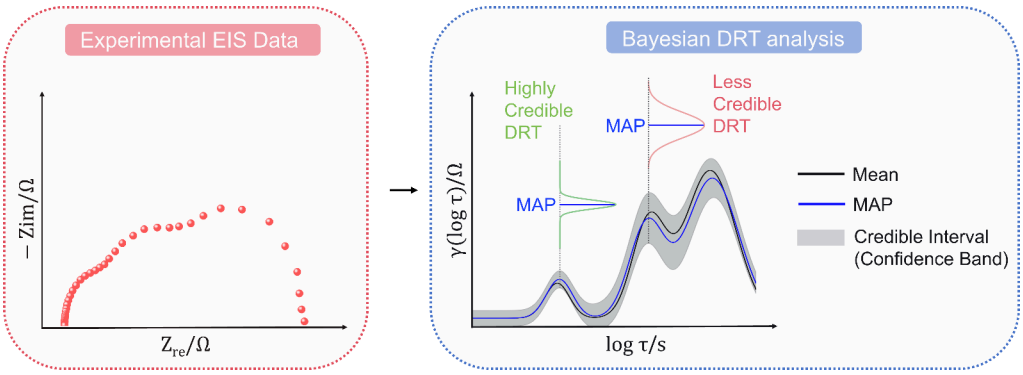


Figure 5. Schematic diagram of Bayesian DRT analysis.

- Users choose their own λ using the "*Custom*" option. This method allows for direct user control and can be a fair choice, particularly for experienced users who can leverage their expertise. However, it is inherently subjective, relying on expert judgment rather than objective criteria.
- The platform incorporates several automated parameter selection methods. These include the L-curve (LC) method and various cross-validation techniques such as generalized cross-validation (GCV), modified GCV (mGCV), and k-fold cross-validation (Kf).[7] When a user selects one of these data-driven options, the platform automatically calculates the optimal λ based on the chosen criterion during the DRT computation.

Users may inspect results interactively by adjusting λ and observing the resolution−smoothness trade-off and/or invoke our automatic λ-selection tools (e.g. mGCV) and then run the DRT calculations. Our platform will show the chosen λ (by manual or automatic selection tools) after calculations.

Options for RBF. In addition to selecting the RBF type, users can adjust its characteristics to exercise granular control over the basis function's morphology, which is a crucial step for optimizing the DRT discretization. The user can adjust the "*FWHM Coefficient*" (full width at half-maximum) or the "*Shape Factor*". By default, the FWHM coefficient (or shape factor) is set to 0.5. Varying the FWHM coefficient (or shape factor) alters the RBFs of DRT peaks: narrower RBFs better resolve closely spaced peaks, potentially increasing the number of discernible features; broader RBFs enhance smoothness, but risk merging adjacent peaks. Accordingly, the FWHM coefficient (or shape factor) should be chosen to maximize the fidelity of the reconstructed impedance to the experimental data (Figure S4 in the Supporting Information).

Distribution of Relaxation Times Run. *Simple DRT Run.* Clicking the "*Simple Run*" button initiates a standard regularized regression to compute a point estimate of the DRT. This method is the fastest computation option and yields a single DRT spectrum. The calculation time depends on the number of data points and the chosen discretization method. The output includes the deconvolved DRT spectrum, the fitted impedance on the Nyquist plot, and the fitted magnitude and phase on the Bode plot (Figure S5 in the Supporting Information).

Following the initial deconvolution, the user can also perform a "*Peak Analysis*" to characterize the DRT peaks when overlapping peaks are suspected. This can be done using two different fitting options: "*Separate Fitting*", which fits individual Gaussian peaks to the DRT, or "*Combine Fitting*", which fits a selected number of Gaussian peaks to the entire recovered DRT at once. For both methods, the user must specify the number of Gaussian peaks to be used for the fitting.

Bayesian DRT Run. Standard regularized regression provides a single point estimate of the DRT but cannot assess the credibility of the result. To address this limitation, a Bayesian approach can be used, which treats all unknown quantities, including the DRT and its parameters, as random variables defined by probability distribution functions. This method contrasts with frequentist approaches like Tikhonov regularization, which seek a single "best" solution.[51,52] The foundation of Bayesian inference is Bayes's theorem, which describes how initial beliefs about parameters, represented by a prior distribution $p(\boldsymbol{x})$, are updated by the likelihood function, $p(\mathbf{Z}_{\text{exp}}|\boldsymbol{x})$), to yield a posterior distribution, $p(\boldsymbol{x}|\mathbf{Z}_{\text{exp}})$. This relationship is expressed as

$$p(\boldsymbol{x}|\mathbf{Z}_{\text{exp}}) = \frac{p(\mathbf{Z}_{\text{exp}}|\boldsymbol{x})p(\boldsymbol{x})}{p(\mathbf{Z}_{\text{exp}})} \tag{3}$$

When applied to DRT, this allows for the incorporation of prior beliefs or assumptions, such as non-negativity and smoothness, through the construction of prior distributions. The result is not a single DRT but a full probability distribution that encapsulates the uncertainty of the estimate given the data and assumptions. While the maximum a posteriori (MAP) estimate from this distribution coincides with the regularized regression solution, a key advantage of the Bayesian framework is its ability to provide credible intervals for the DRT. These intervals quantify the uncertainty in the deconvolved DRT, a crucial feature for assessing the reliability of results, particularly when dealing with noisy or sparse data.

To perform a Bayesian analysis, the user clicks the "*Bayesian DRT Run*" button. This computation is performed using an exact sampler (developed by Paninski et al.),[53] and it returns the MAP estimate, the mean DRT, and the credible interval that quantifies the estimation's uncertainty, features that are not available from the "*Simple Run*". Due to the sampling process involved, the Bayesian run requires more computational time than the simple run. This computation time, as well as the width of the credible interval, can be reduced by selecting the second-order derivative for regularization. Narrower credible intervals indicate higher confidence in the resulting MAP estimate (Figure 5).

The "Number of Sample" parameter determines how many samples are drawn. Increasing the number of samples improves the accuracy of the estimated mean DRT and the credible

https://doi.org/10.1021/acselectrochem.5c00334
ACS Electrochem. 2025, 1, 2680−2689

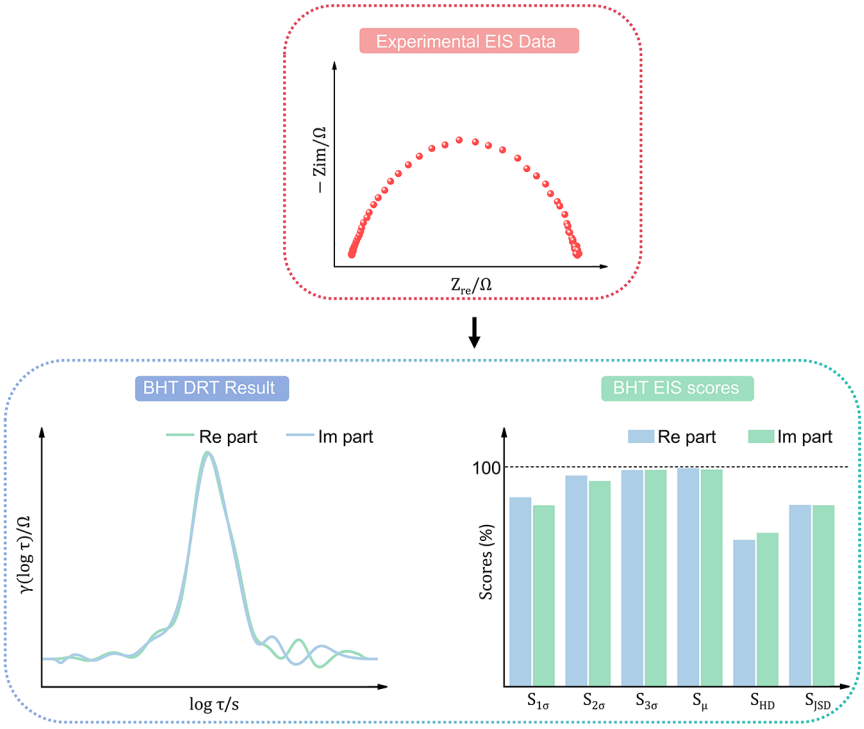


Figure 6. Schematic diagram of BHT analysis with DRT result and EIS scores.

intervals but also increases computation time. To ensure a reasonable estimation, the platform requires a minimum of 1000 samples (Figure S6 in the Supporting Information).

Bayesian Hilbert Transform DRT Run. Assessing the quality and consistency of experimental EIS data is a paramount step in the analysis workflow. For any data from a linear, time-invariant, stable, and causal system, the fundamental Kramers–Kronig (KK) relations must be satisfied.[54−56] These relations are embedded within the Hilbert transform (HT), which mathematically links the real and imaginary parts of the impedance spectrum.[57] While testing the data's compliance with the KK relations or the HT is a crucial validity check, traditional tests are often sensitive to experimental noise, finite frequency ranges, and the data's asymptotic behavior.

To overcome these challenges, the KK test can be reformulated within a Bayesian framework, leading to the BHT. Instead of producing a single, deterministic result, the BHT method treats the measured impedance and its transform as random variables. By modeling the data with uncertainty, the BHT provides a probabilistic estimate of the HT spectrum along with credible intervals, allowing for a more robust assessment of data consistency that accounts for measurement uncertainty.[58] Compared with conventional KK tests, the Bayesian framework delivers a reproducible HT/KK validation protocol with credible intervals, reduces reliance on *ad hoc* thresholds, and pinpoints sources of inconsistency, turning quality control from a single global score into targeted diagnostics.

To perform the Bayesian HT test, the user clicks the "*Bayesian Hilbert Transform Run*" button. This function probabilistically assesses the self-consistency of the experimental EIS data with the KK relations. As shown in Figure 6, the output includes the HT of both the real and imaginary parts of the DRT, along with several EIS quality scores ($S_{1\sigma}$, $S_{2\sigma}$, $S_{3\sigma}$, S_{μ}, S_{HD}, and S_{JSD}). More details are provided in Figure S7 and Table S2 in the Supporting Information.

Data Output and Feedback. After completing the computation, users can view the finalized DRT results in the "*DRT Result Display*" section (Figure 2). The "*Results Output*" section (Figure 2) shows the parameters used in the current calculation. Additionally, download buttons will appear, allowing users to export the DRT result, fitted EIS result, and DRT plot for further analysis or documentation. Users can refer to the "Quick Start Examples" section in the Supporting Information for more details. If users encounter any issues while using DRTtools.com, they are encouraged to seek help or provide feedback by clicking the GitHub icon in the upper-right corner of the webpage, where questions and problems can be reported for prompt support.

■ TUTORIALS FOR MATLAB-BASED DRTtools AND PYTHON-BASED pyDRTtools

In addition to the web platform, offline graphical user interfaces (GUIs) are available for both MATLAB and Python users. The MATLAB-based DRTtools is a GUI for analyzing EIS data with the DRT method. Similarly, pyDRTtools is a user-friendly, Python-based GUI designed for the same purpose. The functionality of both offline tools closely mirrors that of the online DRTtools.com platform, and users can refer to the tutorial for the online version for guidance on their use. It is worth noting that for advanced users, the MATLAB-based DRTtools and pyDRTtools offer greater programmability and flexibility. These platforms are well-suited for users with programming experience who wish to customize more advanced functionalities.

■ CONCLUSION AND OUTLOOK

The DRT provides a less model-dependent analysis of EIS, offering a transparent time-constant representation that can

https://doi.org/10.1021/acselectrochem.5c00334
ACS Electrochem. 2025, 1, 2680−2689

decouple overlapping processes and reveal characteristic timescales. Relative to ECMs, DRT separates processes without prior circuit assumptions, yields cross-condition, multidimensional mechanistic maps, quantifies uncertainty and data consistency, and integrates naturally with modern data-science workflows. DRT and ECMs are complementary: a well-resolved DRT can guide parsimonious ECM selection and parameterization. The DRTtools web platform makes these advanced techniques accessible to a wider community by providing a free and intuitive interface for data import, parameters configuration, and computation. Nevertheless, DRT remains constrained by the ill-posed nature of the underlying deconvolution, necessitating careful regularization to avoid non-physical oscillations or over-smoothing. Automated parameter selection and Bayesian formulations, as implemented in DRTtools, help mitigate these issues. Finally, interpreting the physical meaning of the resulting peaks (especially in complex systems) should be grounded in experimental evidence and informed by domain knowledge, not subjective interpretation.[20]

The DRT is advancing toward greater automation, methodological rigor, and integration with complementary techniques. In practice, this evolution includes (i) mechanistic diagnostics across batteries, fuel cells, electrolyzers, corrosion, and sensing; (ii) operando, multidimensional DRT that explicitly accounts for key variables (e.g., temperature, state-of-charge, pressure, flow); and (iii) uncertainty-quantified outputs that interoperate with ECMs, digital twins, and battery- and energy-management systems (BMS/EMS) for state estimation, fault detection, and control-oriented model selection. A key direction is the combination of DRT with artificial intelligence (AI) and machine learning.[4,59] Vectorized DRT spectra can serve as powerful features for AI models aimed at diagnosing battery state-of-health, predicting lifespans, identifying degradation mechanisms, and classifying materials based on their kinetic fingerprints.[60−62] AI can also be leveraged to automate peak assignment and correlation with specific electrochemical processes, reducing the reliance on manual interpretation.[59] Ultimately, the DRT is designed to become a standard, easily accessible tool in the electrochemical analysis process, integrated into next-generation electrochemical workstations and software suites,[63,64] thereby speeding up materials discovery, improving device performance, and enhancing our understanding of electrochemical systems across all relevant fields.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acselectrochem.5c00334.

Additional information about supplementary figures, tables, and quick start examples (PDF)

■ AUTHOR INFORMATION

Corresponding Authors

Yuhao Wang − *Department of Mechanical and Aerospace Engineering, The Hong Kong University of Science and Technology, Clear Water Bay, Kowloon 999077 Hong Kong S.A.R., China;* orcid.org/0000-0003-2943-434X; Email: yuhao.wang@connect.ust.hk

Francesco Ciucci − *Department of Mechanical and Aerospace Engineering, The Hong Kong University of Science and Technology, Clear Water Bay, Kowloon 999077 Hong Kong S.A.R., China; Electrode Design for Electrochemical Energy Systems, University of Bayreuth, 95448 Bayreuth, Germany; Bavarian Center for Battery Technology (BayBatt), University of Bayreuth, 95447 Bayreuth, Germany;* orcid.org/0000-0003-0614-5537; Email: francesco.ciucci@ust.hk, francesco.ciucci@uni-bayreuth.de

Authors

Zilong Wang − *Department of Mechanical and Aerospace Engineering, The Hong Kong University of Science and Technology, Clear Water Bay, Kowloon 999077 Hong Kong S.A.R., China;* orcid.org/0009-0001-1869-2899

Baptiste Py − *Department of Mechanical and Aerospace Engineering, The Hong Kong University of Science and Technology, Clear Water Bay, Kowloon 999077 Hong Kong S.A.R., China*

Adeleke Maradesa − *Department of Mechanical and Aerospace Engineering, The Hong Kong University of Science and Technology, Clear Water Bay, Kowloon 999077 Hong Kong S.A.R., China*

Jiapeng Liu − *School of Advanced Energy, Sun Yat-Sen University, 518107 Shenzhen, China;* orcid.org/0000-0001-8667-1929

Ting Hei Wan − *Department of Mechanical and Aerospace Engineering, The Hong Kong University of Science and Technology, Clear Water Bay, Kowloon 999077 Hong Kong S.A.R., China;* orcid.org/0000-0002-3860-6847

Mattia Saccoccio − *NitroVolt ApS, 2860 Søborg, Denmark;* orcid.org/0000-0002-5365-3158

Complete contact information is available at: https://pubs.acs.org/10.1021/acselectrochem.5c00334

Author Contributions

#Z.W. and Y.W. contributed equally.

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors gratefully acknowledge the Research Grants Council of Hong Kong for support through the project 16201622, as well as the University of Bayreuth and the Bavarian Center for Battery Technology (BayBatt) for start-up funding. This work was partly funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation, 533115776). Y. Wang acknowledges support from the RGC Junior Research Fellow Scheme (JRFS2526-6S08). B. Py thanks the Hong Kong PhD Fellowship Scheme for its financial support.

■ LIST OF PARAMETERS

τ	Timescale, characteristic relaxation time associated with an electrochemical process
Z	Impedance, defined as the ratio of voltage to current in the frequency domain
Z'	Real part of impedance
$-Z''$	Negative imaginary part of impedance
f	Frequency, excitation frequency (Hz) of the AC perturbation used in EIS
ω	Angular frequency, $\omega = 2\pi f$
R_∞	Ohmic resistance

R_p Total polarization resistance, the integrated resistive contribution of all relaxation processes; equals the area under the DRT

i Imaginary unit, $i^2 = -1$

L_0 Inductive component, lumped inductance modeling wiring/fixture inductive effects appearing at high frequencies

λ Regularization parameter, controlling the trade-off between data fidelity and smoothness in DRT deconvolution

$\boldsymbol{L}$ Derivative/differentiation matrix, linear operator penalizing first/second derivatives of the discretized DRT to enforce smoothness

$\boldsymbol{x}$ Vector of discretized DRT at log τ

$\mathbf{A}$ Discretization matrix, forward model operator linking the discretized DRT to impedance

$\mathbf{Z}_{\text{exp}}$ Experimental impedance

$\|\cdot\|$ Euclidean norm

$p(\boldsymbol{x})$ Prior distribution, probabilistic encoding of assumptions (e.g., smoothness, non-negativity) over the discretized DRT

$p(\mathbf{Z}_{\text{exp}}|\boldsymbol{x})$ Likelihood function of experiment, probability distribution model linking a candidate DRT to observed impedance

$p(\boldsymbol{x}|\mathbf{Z}_{\text{exp}})$ Posterior distribution, Bayesian update combining prior and likelihood to quantify uncertainty in the DRT

$S_{k\sigma}$ Score based on different residuals, the fraction of frequency points whose residuals (experimental minus BHT mean prediction) fall within the $k\sigma$ posterior credible band (k = 1,2,3), averaged over frequency.

S_{μ} Score based on the estimated mean discrepancy, agreement of the posterior means obtained from the two inference pathways (real → imaginary and imaginary → real)

S_{HD} Score based on the Hellinger distance, defined as 1 minus the average Hellinger distance between the two Gaussian posteriors

S_{JSD} Score based on the Jensen−Shannon discrepancy, defined as 1 minus the normalized average JSD (a symmetrical between the two posteriors)

■ REFERENCES

(1) Ciucci, F. Modeling electrochemical impedance spectroscopy. *Curr. Opin. Electrochem.* **2019**, *13*, 132−139.

(2) Orazem, M. E.; Tribollet, B. *Electrochemical impedance spectroscopy*; Wiley: 2008; pp 383−389. .

(3) Lasia, A. Electrochemical impedance spectroscopy and its applications. *Modern aspects of electrochemistry*; Springer: 2002; pp 143−248.

(4) Maradesa, A.; Py, B.; Huang, J.; Lu, Y.; Iurilli, P.; Mrozinski, A.; Law, H. M.; Wang, Y.; Wang, Z.; Li, J.; Xu, S.; Meyer, Q.; Liu, J.; Brivio, C.; Gavrilyuk, A.; Kobayashi, K.; Bertei, A.; Williams, N. J.; Zhao, C.; Danzer, M.; Zic, M.; Wu, P.; Yrjänä, V.; Pereverzyev, S.; Chen, Y.; Weber, A.; Kalinin, S. V.; Schmidt, J. P.; Tsur, Y.; Boukamp, B. A.; Zhang, Q.; Gaberšček, M.; O'Hayre, R.; Ciucci, F. Advancing electrochemical impedance analysis through innovations in the distribution of relaxation times method. *Joule* **2024**, *8* (7), 1958−1981.

(5) Quattrocchi, E.; Wan, T. H.; Belotti, A.; Kim, D.; Pepe, S.; Kalinin, S. V.; Ahmadi, M.; Ciucci, F. The deep-DRT: A deep neural network approach to deconvolve the distribution of relaxation times from multidimensional electrochemical impedance spectroscopy data. *Electrochimica Acta* **2021**, *392*, No. 139010.

(6) Wan, T. H.; Saccoccio, M.; Chen, C.; Ciucci, F. Influence of the Discretization Methods on the Distribution of Relaxation Times Deconvolution: Implementing Radial Basis Functions with DRTtools. *Electrochimica Acta* **2015**, *184*, 483−499.

(7) Maradesa, A.; Py, B.; Wan, T. H.; Effat, M. B.; Ciucci, F. Selecting the Regularization Parameter in the Distribution of Relaxation Times. *J. Electrochem. Soc.* **2023**, *170* (3), No. 030502.

(8) Quattrocchi, E.; Py, B.; Maradesa, A.; Meyer, Q.; Zhao, C.; Ciucci, F. Deconvolution of electrochemical impedance spectroscopy data using the deep-neural-network-enhanced distribution of relaxation times. *Electrochimica Acta* **2023**, *439*, No. 141499.

(9) Maradesa, A.; Py, B.; Ciucci, F. Probabilistic deconvolution of the distribution of relaxation times from multiple electrochemical impedance spectra. *J. Power Sources* **2024**, *621*, No. 235236.

(10) Wang, Z. Distribution of relaxation times for impedance analysis in batteries and fuel cells. *Nat. Rev. Clean Technol.* **2025**, *1* (7), 453−453.

(11) Schmidt, J. P.; Berg, P.; Schönleber, M.; Weber, A.; Ivers-Tiffée, E. The distribution of relaxation times as basis for generalized time-domain models for Li-ion batteries. *J. Power Sources* **2013**, *221*, 70−77.

(12) Avioz Cohen, G.; Gelman, D.; Tsur, Y. Development of a Typical Distribution Function of Relaxation Times Model for Polymer Electrolyte Membrane Fuel Cells and Quantifying the Resistance to Proton Conduction within the Catalyst Layer. *J. Phys. Chem. C* **2021**, *125* (22), 11867−11874.

(13) Attias, R.; Bhowmick, S.; Tsur, Y. Distribution function of relaxation times: An alternative to classical methods for evaluating the reaction kinetics of oxygen evolution reaction. *Chem. Eng. J.* **2023**, *476*, No. 146708.

(14) Lu, Y.; Zhao, C.-Z.; Huang, J.-Q.; Zhang, Q. The timescale identification decoupling complicated kinetic processes in lithium batteries. *Joule* **2022**, *6* (6), 1172−1198.

(15) Boukamp, B. A. Fourier transform distribution function of relaxation times; application and limitations. *Electrochimica Acta* **2015**, *154*, 35−46.

(16) Ivers-Tiffée, E.; Weber, A. Evaluation of electrochemical impedance spectra by the distribution of relaxation times. *J. Ceram. Soc. Japan* **2017**, *125* (4), 193−201.

(17) Dierickx, S.; Weber, A.; Ivers-Tiffée, E. How the distribution of relaxation times enhances complex equivalent circuit models for fuel cells. *Electrochimica Acta* **2020**, *355*, No. 136764.

(18) Huang, J.; Sullivan, N. P.; Zakutayev, A.; O'Hayre, R. How reliable is distribution of relaxation times (DRT) analysis? A dual regression-classification perspective on DRT estimation, interpretation, and accuracy. *Electrochimica Acta* **2023**, *443*, No. 141879.

(19) Li, X.; Ahmadi, M.; Collins, L.; Kalinin, S. V. Deconvolving distribution of relaxation times, resistances and inductance from electrochemical impedance spectroscopy via statistical model selection: Exploiting structural-sparsity regularization and data-driven parameter tuning. *Electrochimica Acta* **2019**, *313*, 570−583.

(20) Chen, J.; Quattrocchi, E.; Ciucci, F.; Chen, Y. Charging processes in lithium-oxygen batteries unraveled through the lens of the distribution of relaxation times. *Chem* **2023**, *9* (8), 2267−2281.

(21) Schlüter, N.; Ernst, S.; Schröder, U. Finding the Optimal Regularization Parameter in Distribution of Relaxation Times Analysis. *ChemElectroChem* **2019**, *6* (24), 6027−6037.

(22) Paul, T.; Chi, P. W.; Wu, P. M.; Wu, M. K. Computation of distribution of relaxation times by Tikhonov regularization for Li ion batteries: usage of L-curve method. *Sci. Rep.* **2021**, *11* (1), No. 12624.

(23) Goldammer, E.; Kowal, J. Determination of the Distribution of Relaxation Times by Means of Pulse Evaluation for Offline and Online Diagnosis of Lithium-Ion Batteries. *Batteries* **2021**, *7*, 36.

(24) Semerukhin, D. Y.; Kubarkov, A. V.; Sergeyev, V. G.; Semenikhin, O. A.; Antipov, E. V. Analysis of the Distribution of Relaxation Times (DRT) Responses of Li-Ion Cells as a Function of Their Preparation Conditions. *Electrochimica Acta* **2024**, *486*, No. 144092.

https://doi.org/10.1021/acselectrochem.5c00334
ACS Electrochem. 2025, 1, 2680−2689

(25) Liang, M.; Wang, Y.; Song, Y.; Guan, D.; Wu, J.; Chen, P.; Maradesa, A.; Xu, M.; Yang, G.; Zhou, W.; Wang, W.; Ran, R.; Ciucci, F.; Shao, Z. High-temperature water oxidation activity of a perovskite-based nanocomposite towards application as air electrode in reversible protonic ceramic cells. *Appl. Catal. B: Environ.* **2023**, *331*, No. 122682.

(26) Huang, H.; Zhang, Z.; Xiao, C.; Liu, J.; Li, Z.; Jiang, Y.; Wei, L.; Zhao, T.; Ciucci, F.; Zeng, L. Water Management Fault Diagnosis by Operando Distribution of Relaxation Times Analysis for Anion Exchange Membrane Fuel Cells. *Adv. Sci.* **2025**, *12*, No. 2505304.

(27) Boukamp, B. A.; Rolle, A. Use of a distribution function of relaxation times (DFRT) in impedance analysis of SOFC electrodes. *Solid State Ionics* **2018**, *314*, 103−111.

(28) Reshetenko, T.; Kulikovsky, A. Understanding the distribution of relaxation times of a low−Pt PEM fuel cell. *Electrochimica Acta* **2021**, *391*, No. 138954.

(29) Choi, J.; Jang, D.; Kim, M.; Ha, J.; Ahn, H.; Kim, W. B. Multifunctional perovskite oxide $Pr_{0.6}Sr_{0.4}Mn_{0.2}Fe_{0.7}Ni_{0.1}O_{3-\delta}$ as an efficient quasi-symmetric electrode for solid oxide fuel/electrolysis cells. *J. Mater. Chem. A* **2024**, *12* (21), 12619−12627.

(30) Zhu, H.; Liu, Y.; Guo, W.; Zheng, J.; Zheng, Y.; Cherevko, S.; Tang, C.; Zhang, Q. Electrochemical impedance spectroscopy analysis to accelerate electrocatalytic system innovation. *Sci. China Chem.* **2024**, *67* (12), 3964−3975.

(31) Wang, Y.; Qiao, S.; Xie, J.; Wang, Y.; Hao, L.; Wang, J.; Ke, W. DRT analysis on EIS data to identify aerobic-anaerobic corrosion evolution of low-carbon steel under deep geological disposal condition. *Corrosion Sci.* **2025**, *255*, No. 113118.

(32) Patel, R. D.; Chaudhary, K. J.; Purohit, D.; Prochowicz, D.; Akin, S.; Kalam, A.; Khanna, S.; Pandey, S. V.; Yadav, P. Dual DRT-based deconvolution of electrochemical signatures in perovskite solar cells. *J. Energy Chem.* **2025**, *109*, 975−984.

(33) Piccioni, A.; Vecchi, P.; Vecchi, L.; Grandi, S.; Caramori, S.; Mazzaro, R.; Pasquini, L. Distribution of Relaxation Times Based on Lasso Regression: A Tool for High-Resolution Analysis of IMPS Data in Photoelectrochemical Systems. *J. Phys. Chem. C* **2023**, *127* (17), 7957−7964.

(34) Osipenko, M. A.; Karczewski, J.; Dominów, M.; Prześniak-Welenc, M.; Makarava, I. V.; Kurilo, I.; Kharytonau, D. S.; Ryl, J. Multisine impedimetric monitoring with an in-depth distribution of relaxation times analysis of WE43 and AZ31 magnesium alloys corrosion. *Measurement* **2023**, *222*, No. 113683.

(35) Ramírez-Chavarría, R. G.; Sánchez-Pérez, C.; Matatagui, D. *Analysis of impedance spectroscopy measurements of biological tissue using the distribution of relaxation times method*; SCITEPRESS, 2017; pp 224−228.

(36) Ramírez-Chavarría, R. G.; Sánchez-Pérez, C.; Romero-Ornelas, L.; Ramón-Gallegos, E. Time-constant-domain spectroscopy: an impedance-based method for sensing biological cells in suspension. *IEEE Sensors J.* **2021**, *21* (1), 185−192.

(37) Van Haeverbeke, M.; De Baets, B.; Stock, M. Evaluating the potential of Distribution of Relaxation Times analysis for plant agriculture. *Comput. Electron. Agric.* **2023**, *213*, No. 108249.

(38) Li, Z.; Feng, H.; Jia, B.; Zuo, Z.; Mei, B.-A. Identification of thermal process of electric double layer capacitor systems by distribution of relaxation times analysis for thermal impedance spectroscopy. *J. Energy Storage* **2023**, *64*, No. 107152.

(39) Ramírez-Chavarría, R. G.; Castillo-Villanueva, E.; Alvarez-Serna, B. E.; Carrillo-Reyes, J.; Torres, L.; Ramírez-Zamora, R. M.; Buitrón, G.; Alvarez-Icaza, L. Automatic Analysis of Isothermal Amplification via Impedance Time-Constant-Domain Spectroscopy: A SARS-CoV-2 Case Study. *Chemosensors* **2023**, *11*, 230.

(40) Shi, F.; Kolb, J. F. Enhanced resolution impedimetric analysis of cell responses from the distribution of relaxation times. *Biosens. Bioelectron.* **2020**, *157*, No. 112149.

(41) Ramírez-Chavarría, R. G.; Sánchez-Pérez, C.; Matatagui, D.; Qureshi, N.; Pérez-García, A.; Hernández-Ruíz, J. Ex-vivo biological tissue differentiation by the Distribution of Relaxation Times method applied to Electrical Impedance Spectroscopy. *Electrochimica Acta* **2018**, *276*, 214−222.

(42) Macdonald, J. R.; Johnson, W. B. Fundamentals of Impedance Spectroscopy. *Impedance Spectrosc.* **2005**, 1−26.

(43) Saccoccio, M.; Wan, T. H.; Chen, C.; Ciucci, F. Optimal Regularization in Distribution of Relaxation Times applied to Electrochemical Impedance Spectroscopy: Ridge and Lasso Regression Methods - A Theoretical and Experimental Study. *Electrochimica Acta* **2014**, *147*, 470−482.

(44) Clematis, D.; Ferrari, T.; Bertei, A.; Asensio, A. M.; Carpanese, M. P.; Nicolella, C.; Barbucci, A. On the stabilization and extension of the distribution of relaxation times analysis. *Electrochimica Acta* **2021**, *391*, No. 138916.

(45) Py, B.; Wang, Z.; Wang, Y.; Ciucci, F. Entropy-based regularized regression for advanced distribution of relaxation times deconvolution. *J. Power Sources* **2025**, *644*, No. 236910.

(46) Liu, J.; Ciucci, F. The Gaussian process distribution of relaxation times: A machine learning tool for the analysis and prediction of electrochemical impedance spectroscopy data. *Electrochimica Acta* **2020**, *331*, No. 135316.

(47) Gavrilyuk, A. L.; Osinkin, D. A.; Bronin, D. I. The use of Tikhonov regularization method for calculating the distribution function of relaxation times in impedance spectroscopy. *Russian J. Electrochem.* **2017**, *53* (6), 575−588.

(48) Py, B.; Ciucci, F. Beyond Ridge Regression: Enhancing Distribution of Relaxation Times Deconvolution. *J. Electrochem. Soc.* **2024**, *171* (6), No. 060529.

(49) Maradesa, A.; Py, B.; Quattrocchi, E.; Ciucci, F. The probabilistic deconvolution of the distribution of relaxation times with finite Gaussian processes. *Electrochimica Acta* **2022**, *413*, No. 140119.

(50) Py, B.; Maradesa, A.; Ciucci, F. Gaussian processes for the analysis of electrochemical impedance spectroscopy data: Prediction, filtering, and active learning. *Electrochimica Acta* **2023**, *439*, No. 141688.

(51) Ciucci, F.; Chen, C. Analysis of Electrochemical Impedance Spectroscopy Data Using the Distribution of Relaxation Times: A Bayesian and Hierarchical Bayesian Approach. *Electrochimica Acta* **2015**, *167*, 439−454.

(52) Effat, M. B.; Ciucci, F. Bayesian and Hierarchical Bayesian Based Regularization for Deconvolving the Distribution of Relaxation Times from Electrochemical Impedance Spectroscopy Data. *Electrochimica Acta* **2017**, *247*, 1117−1129.

(53) Pakman, A.; Paninski, L. Auxiliary-variable exact Hamiltonian Monte Carlo samplers for binary distributions. *Advances in neural information processing systems 26* (NIPS 2013); NeurIPS Proceedings, 2013.

(54) Schönleber, M.; Klotz, D.; Ivers-Tiffée, E. A Method for Improving the Robustness of linear Kramers-Kronig Validity Tests. *Electrochimica Acta* **2014**, *131*, 20−27.

(55) Boukamp, B. A. A Linear Kronig-Kramers Transform Test for Immittance Data Validation. *J. Electrochem. Soc.* **1995**, *142* (6), 1885.

(56) Agarwal, P.; Orazem, M. E.; Garcia-Rubio, L. H. Application of Measurement Models to Impedance Spectroscopy: III. Evaluation of Consistency with the Kramers-Kronig Relations. *J. Electrochem. Soc.* **1995**, *142* (12), 4159.

(57) Ciucci, F. The Gaussian Process Hilbert Transform (GP-HT): Testing the Consistency of Electrochemical Impedance Spectroscopy Data. *J. Electrochem. Soc.* **2020**, *167* (12), No. 126503.

(58) Liu, J.; Wan, T. H.; Ciucci, F. A Bayesian view on the Hilbert transform and the Kramers-Kronig transform of electrochemical impedance data: Probabilistic estimates and quality scores. *Electrochimica Acta* **2020**, *357*, No. 136864.

(59) Liu, X.; Yan, Z.; Wu, J.; Huang, J.; Zheng, Y.; Sullivan, N. P.; O'Hayre, R.; Zhong, Z.; Pan, Z. Prediction of impedance responses of protonic ceramic cells using artificial neural network tuned with the distribution of relaxation times. *J. Energy Chem.* **2023**, *78*, 582−588.

(60) Li, H.; Jin, Y.; Yu, D. A DRT based Lithium Battery Multi-Time Scale Model Parameter Identification and SOC Estimation. Presented

at the International Conference on Sensing, Measurement & Data Analytics in the Era of Artificial Intelligence (ICSMD 2023), Xi'an, China, 2−4 Nov 2023; 2023; pp 1−7.
(61) Chen, X.; Li, Q.; Shao, B.; Dou, W.; Lai, C.; Lu, T.; Xie, J. Accurately estimating internal temperature of lithium-ion batteries based on the distribution of relaxation time and data-driven. *Energy* **2025**, *320*, No. 135493.
(62) Cai, X.; Zhang, C.; Ruan, H.; Chen, Z.; Zhang, L.; Sauer, D. U.; Li, W. Cross-Scale Decoupling Kinetic Processes in Lithium-Ion Batteries Using the Multi-Dimensional Distribution of Relaxation Time. *Adv. Sci.* **2024**, *11* (44), No. 2406934.
(63) *ZView for Windows*, version 4.1; Scribner Associates, Inc., 2025.
(64) *RelaxIS 3: Impedance Spectrum Analysis*; rhd Instruments GmbH & Co. KG, 2025.

https://doi.org/10.1021/acselectrochem.5c00334
ACS Electrochem. 2025, 1, 2680−2689